FUNDACJA
QUANTUM AI

Q Poland

Quantum HACKATHON

16-18.10.2026

Strategic Partners





Production Scheduling Optimization — the ERP Challenge

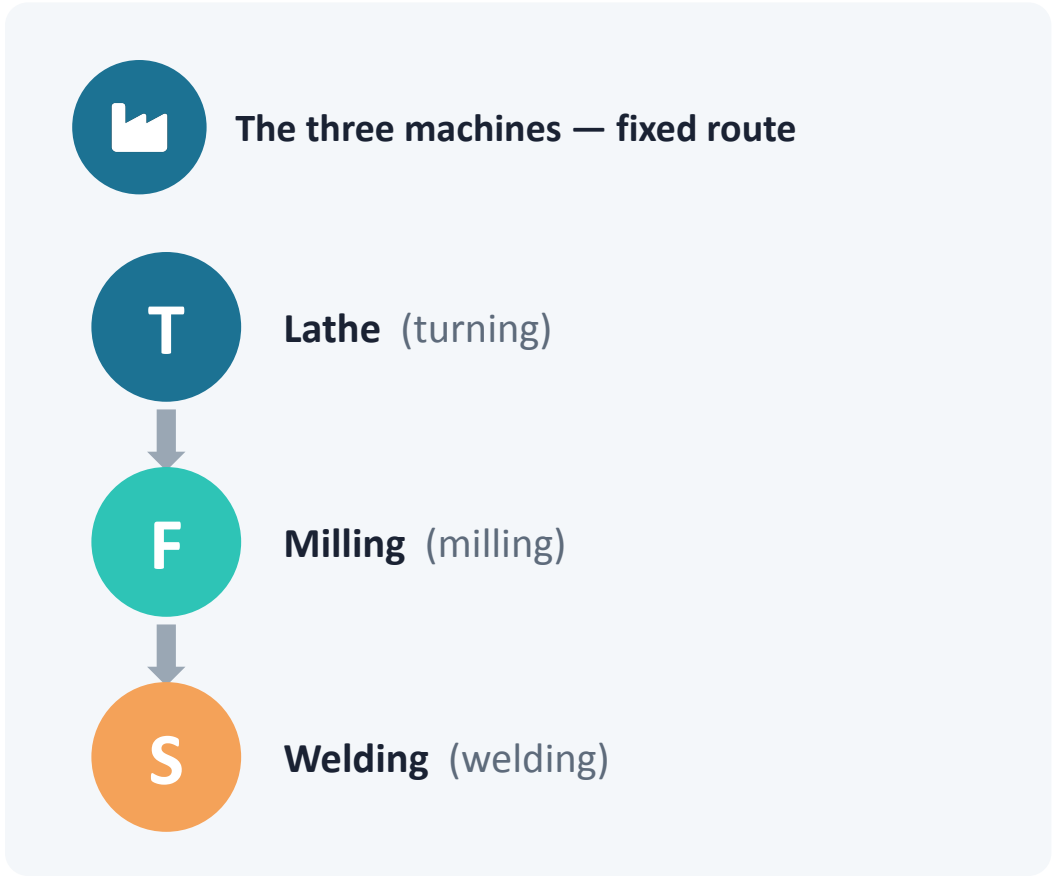
Solve a real factory scheduling problem on a quantum simulator and a QUBO solver.

Friday, 4 PM — an urgent order lands

MetalTech Ltd. makes metal parts to order. The production site has three machines, and every part must pass through them in a fixed order.

Just before the weekend, a rush order arrives: **three parts (Z1, Z2, Z3)**, each with different times on each machine — all needed as fast as possible.

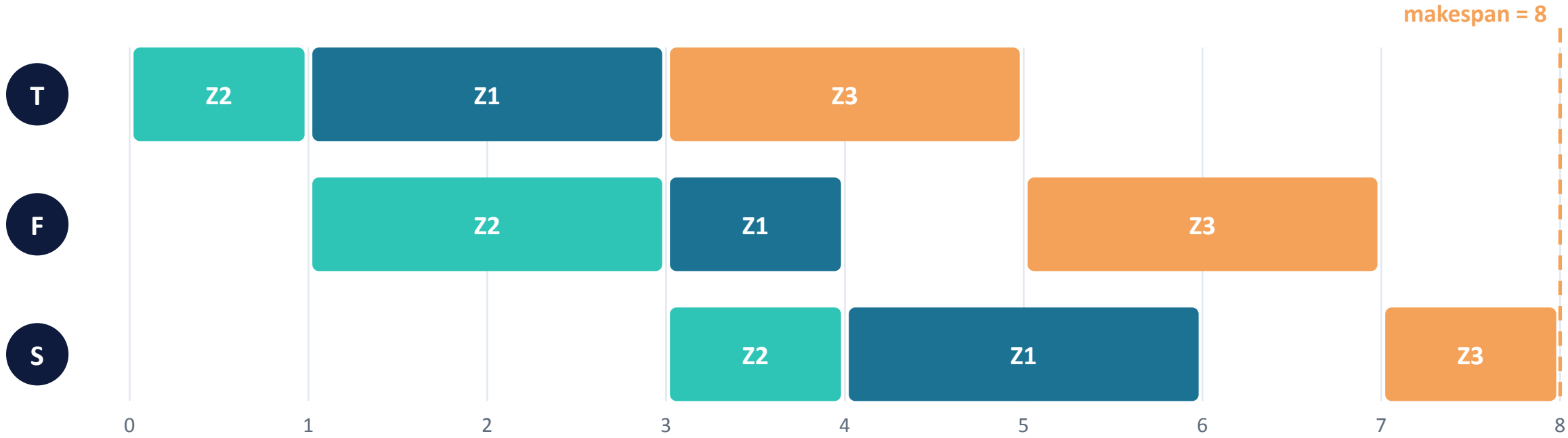
Mr. Zenon, the production manager, must sequence the work so the whole batch finishes as early as possible.



The goal: minimize the makespan

Schedule all operations so the last one finishes as early as possible — respecting three rules:

- Each operation starts once
- No machine runs two ops at once
- Order T → F → S per job



Example optimal schedule — full 3×3 instance (jobs Z1 / Z2 / Z3)

How the challenge works

STAGE 1

Mandatory baseline

A fixed problem everyone solves. Today it's roughly one good AI prompt — so everyone lands at a similar level and **we don't differentiate on it.**

Your entry ticket.

STAGE 2

Your own extension

Build your own, more ambitious problem on top of the baseline. **This is where the points are** — depth, rigor, and your own ideas.

Where you win.

"The baseline proves you can do it; the extension proves you understand it — and can take it somewhere new."

Stage 1 — the two instances you must deliver

INSTANCE A

2 jobs × 2 machines

Machines	**T → F**
Solved on	**Quantum algorithm · hardware or simulator**
Returns	**Feasible schedule → schedule_2x2.json**

INSTANCE B

3 jobs × 3 machines

Machines	**T → F → S**
Solved on	**Any QUBO solver · quantum / hybrid preferred**
Returns	**Feasible schedule → schedule_3x3.json**

Both are required. No working baseline → not assessed further. Each instance must return a feasible schedule, written to JSON (format in the challenge description).

Stage 2 — extension directions (combine freely)

Optimization / science

Examples:

- Qubit reduction — shrink the 3×3 QUBO without losing correctness
- VeloxQ in action — find a compelling application of VeloxQ solver in your solution
- Compare QAOA vs VQE vs annealing; error mitigation
- Scaling & variance — quality vs. size, statistics over runs

Functional / business

Examples:

- Due dates & tardiness; sequence-dependent setup times
- Disruptions & machine outages, handled robustly
- Rolling planning from a partially-finished shop
- “Should MetalTech buy a second machine?” — model it

What you'll deliver

1

Code repository — public GitHub, MIT license

Both baselines + your extension, a README (how to run, how AI was used) and the two schedule JSON files.

2

Development log & Project Design

Dated log: how it grew, problems, decisions & why. Project Design: your data model and method.

3

Demo video — max 5 min

A link is fine (unlisted OK). Show it running; walk through the key code and your own contributions.

4

Live presentation — 3 min + ~2 min Q&A

Online, with the jury — the real test of whether you understand your own code.

Remember to

Lock in your team

Declare your team in the #lock-in-teams channel on Discord — before the submission deadline.

Read the challenge description

The exact rules, deliverables and scoring are there. Check the details before you build.

Get VeloxQ access

Email support@veloxq.com (cc contact@qaif.org), subject “QPoland QUANTUM HACKATHON 2026”.

Submit on time

Before 18.10.2026, 15:00 UTC+2 — via the Google Form. Only the repo state at the deadline counts.

How we score — and the AI rule



Using AI is fully allowed

— a natural part of the process, and the shared starting point.

Understand your solution to the core.
You must be able to explain it and take full responsibility for it. As you investigate and learn how it works, record that in the Development Log too. Describe how AI was used in your README.

What earns points (max 30)

- QUBO formulation correctness **5 pts**
- Solution quality (makespan) **5 pts**
- Creativity & complexity of the extension **5 pts**
- Development log **3 pts**
- Project Design **3 pts**
- Demo video **3 pts**
- Live presentation & answers **3 pts**
- Consistency: log ↔ code ↔ presentation **3 pts**

Timeline — one weekend

The event runs over one weekend — Friday to Sunday (times in UTC+2).

Day 1 · Fri (18:00)

Kickoff

1

Form your team, lock it in on Discord and start working.

Sat–Sun · till 15:00

Build & submit

2

Complete Instances A and B, build your extension, submit by 15:00.

Sun · 18:00

Live presentations

3

Live presentations — 3 min + ~2 min Q&A per team.

Prizes



2000 PLN

Overall winner

The strongest overall submission, across all judging criteria.



1000 PLN

Best use of VeloxQ

The team that finds the most creative and compelling use for VeloxQ.



Questions? Ask the organizers and mentors anytime on Discord.

Good luck!